

Functional Programming

Data Wrangling in R

Functional Programming

“R, at its heart, is a functional programming (FP) language. This means that it provides many tools for the creation and manipulation of functions.” - [Hadley Wickham](#)

Don't need to write for-loops! - check this [video](#).

Iterative work

Allows you to flexibly iterate functions to multiple elements of a data object!

Useful when you want to apply a function to:

- lots of columns in a tibble
- multiple tibbles
- multiple data files

base R **apply** functions

Works really simply for all columns, but not a tibble.

```
sapply(mtcars, FUN = round)
```

```
##      mpg  cyl  disp  hp  drat  wt  qsec  vs  am  gear  carb
## [1,]  21    6  160 110    4    3   16    0    1     4     4
## [2,]  21    6  160 110    4    3   17    0    1     4     4
## [3,]  23    4  108  93    4    2   19    1    1     4     1
## [4,]  21    6  258 110    3    3   19    1    0     3     1
## [5,]  19    8  360 175    3    3   17    0    0     3     2
## [6,]  18    6  225 105    3    3   20    1    0     3     1
## [7,]  14    8  360 245    3    4   16    0    0     3     4
## [8,]  24    4  147  62    4    3   20    1    0     4     2
## [9,]  23    4  141  95    4    3   23    1    0     4     2
## [10,] 19    6  168 123    4    3   18    1    0     4     4
## [11,] 18    6  168 123    4    3   19    1    0     4     4
## [12,] 16    8  276 180    3    4   17    0    0     3     3
## [13,] 17    8  276 180    3    4   18    0    0     3     3
## [14,] 15    8  276 180    3    4   18    0    0     3     3
## [15,] 10    8  472 205    3    5   18    0    0     3     4
## [16,] 10    8  460 215    3    5   18    0    0     3     4
## [17,] 15    8  440 230    3    5   17    0    0     3     4
## [18,] 32    4    79  66    4    2   19    1    1     4     1
## [19,] 30    4    76  52    5    2   19    1    1     4     2
## [20,] 34    4    71  65    4    2   20    1    1     4     1
## [21,] 22    4   120  97    4    2   20    1    0     3     1
## [22,] 16    8  318 150    3    4   17    0    0     3     2
## [23,] 15    8  304 150    3    3   17    0    0     3     2
```

Working across multiple columns

`across` allows us to perform functions on specific columns more easily. Use with `mutate` or `summarize`.

```
mtcars %>%  
  mutate(across(  
    c(mpg, disp, hp, drat),  
    round # no parentheses if no arguments  
  ))
```

	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
## Mazda RX4	21	6	160	110	4	2.620	16.46	0	1	4	4
## Mazda RX4 Wag	21	6	160	110	4	2.875	17.02	0	1	4	4
## Datsun 710	23	4	108	93	4	2.320	18.61	1	1	4	1
## Hornet 4 Drive	21	6	258	110	3	3.215	19.44	1	0	3	1
## Hornet Sportabout	19	8	360	175	3	3.440	17.02	0	0	3	2
## Valiant	18	6	225	105	3	3.460	20.22	1	0	3	1
## Duster 360	14	8	360	245	3	3.570	15.84	0	0	3	4
## Merc 240D	24	4	147	62	4	3.190	20.00	1	0	4	2
## Merc 230	23	4	141	95	4	3.150	22.90	1	0	4	2
## Merc 280	19	6	168	123	4	3.440	18.30	1	0	4	4
## Merc 280C	18	6	168	123	4	3.440	18.90	1	0	4	4
## Merc 450SE	16	8	276	180	3	4.070	17.40	0	0	3	3
## Merc 450SL	17	8	276	180	3	3.730	17.60	0	0	3	3
## Merc 450SLC	15	8	276	180	3	3.780	18.00	0	0	3	3
## Cadillac Fleetwood	10	8	472	205	3	5.250	17.98	0	0	3	4
## Lincoln Continental	10	8	460	215	3	5.424	17.82	0	0	3	4
## Chrysler Imperial	15	8	440	230	3	5.345	17.42	0	0	3	4
## Fiat 128	32	4	79	66	4	2.200	19.47	1	1	4	1

Using `across` with arguments

If you want to use arguments, then you need to use the `~` and `.` (or `.x`) as a placeholder for what you will be passing into the function.

See [documentation for purrr-shortcuts](#) and [a comparison of `.` and `.x`](#).

```
mtcars %>%  
  mutate(across(  
    c(mpg, disp, hp, drat),  
    ~ round(., digits = -1) # note ~ and .  
  ))
```

##	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
## Mazda RX4	20	6	160	110	0	2.620	16.46	0	1	4	4
## Mazda RX4 Wag	20	6	160	110	0	2.875	17.02	0	1	4	4
## Datsun 710	20	4	110	90	0	2.320	18.61	1	1	4	1
## Hornet 4 Drive	20	6	260	110	0	3.215	19.44	1	0	3	1
## Hornet Sportabout	20	8	360	180	0	3.440	17.02	0	0	3	2
## Valiant	20	6	220	100	0	3.460	20.22	1	0	3	1
## Duster 360	10	8	360	240	0	3.570	15.84	0	0	3	4
## Merc 240D	20	4	150	60	0	3.190	20.00	1	0	4	2
## Merc 230	20	4	140	100	0	3.150	22.90	1	0	4	2
## Merc 280	20	6	170	120	0	3.440	18.30	1	0	4	4
## Merc 280C	20	6	170	120	0	3.440	18.90	1	0	4	4
## Merc 450SE	20	8	280	180	0	4.070	17.40	0	0	3	3
## Merc 450SL	20	8	280	180	0	3.730	17.60	0	0	3	3
## Merc 450SLC	20	8	280	180	0	3.780	18.00	0	0	3	3
## Cadillac Fleetwood	10	8	470	200	0	5.250	17.98	0	0	3	4
## Lincoln Continental	10	8	460	220	0	5.424	17.02	0	0	3	4

Writing functions

Why write your own functions?

- Cut down on repetitive code (easier to fix things!)
- Organize code into manageable chunks
- Avoid running code unintentionally

Functions in R

To create a function we can use the `function` function and specify what input the function will take and what it will do to it.

```
my_function <- function(x){x + 1}  
my_function
```

```
## function (x)  
## {  
##     x + 1  
## }  
## <environment: 0x1450a4e80>
```

```
my_data <- c(2,3,4)  
my_function(x = my_data)
```

```
## [1] 3 4 5
```

Shortcut for functions

Alternatively we can use `\(x)`. See this [link about function shortcuts](#).

```
my_function <- \(x){x + 1}  
my_function
```

```
## function (x)  
## {  
##     x + 1  
## }  
## <environment: 0x1450a4e80>
```

```
my_function(x = my_data)
```

```
## [1] 3 4 5
```

purrr is a super helpful package for iteration!

“Designed to make your functions purrr.”

`dplyr` is designed for data frames `purrr` is designed for vectors

The `purrr` package can be very helpful!

- <https://purrr.tidyverse.org/>
- <https://github.com/rstudio/cheatsheets/raw/master/purrr.pdf>
- <https://jennybc.github.io/purrr-tutorial/>

purrr functions: `map` and `modify`

applies function to each element of an vector or object

- `map` will output a list
- `map_dbl` will output a vector
- `modify` will output the same object type

purrr functions: map and modify

Apply sum to every column

```
mtcars %>% map(sum)
```

```
## $mpg
## [1] 642.9
##
## $cyl
## [1] 198
##
## $disp
## [1] 7383.1
##
## $hp
## [1] 4694
##
## $drat
## [1] 115.09
##
## $wt
## [1] 102.952
##
## $qsec
## [1] 571.16
##
## $vs
## [1] 14
##
```

purrr functions: map and modify

Apply sum to every column - nicer output

```
mtcars %>% map_dbl(sum)
```

```
##      mpg      cyl    disp    hp    drat    wt     qsec     vs
## 642.900  198.000 7383.100 4694.000 115.090 102.952 571.160 14.000
##      am     gear    carb
##  13.000  118.000   90.000
```

purrr functions: map and modify

Apply round to every column - get a data.frame back out

```
mtcars %>% modify(round)
```

##		mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
##	1	21	6	160	110	4	3	16	0	1	4	4
##	2	21	6	160	110	4	3	17	0	1	4	4
##	3	23	4	108	93	4	2	19	1	1	4	1
##	4	21	6	258	110	3	3	19	1	0	3	1
##	5	19	8	360	175	3	3	17	0	0	3	2
##	6	18	6	225	105	3	3	20	1	0	3	1
##	7	14	8	360	245	3	4	16	0	0	3	4
##	8	24	4	147	62	4	3	20	1	0	4	2
##	9	23	4	141	95	4	3	23	1	0	4	2
##	10	19	6	168	123	4	3	18	1	0	4	4
##	11	18	6	168	123	4	3	19	1	0	4	4
##	12	16	8	276	180	3	4	17	0	0	3	3
##	13	17	8	276	180	3	4	18	0	0	3	3
##	14	15	8	276	180	3	4	18	0	0	3	3
##	15	10	8	472	205	3	5	18	0	0	3	4
##	16	10	8	460	215	3	5	18	0	0	3	4
##	17	15	8	440	230	3	5	17	0	0	3	4
##	18	32	4	79	66	4	2	19	1	1	4	1
##	19	30	4	76	52	5	2	19	1	1	4	2
##	20	34	4	71	65	4	2	20	1	1	4	1
##	21	22	4	120	97	4	2	20	1	0	3	1
##	22	16	8	318	150	3	4	17	0	0	3	2
##	23	15	8	304	150	3	3	17	0	0	3	2

Anonymous functions - “functions on the fly”

You can supply a custom function without naming it. Useful if you only need to use it once!

```
mtcars %>% modify\(x) x * 100)
```

##		mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
##	1	2100	600	16000	11000	390	262.0	1646	0	100	400	400
##	2	2100	600	16000	11000	390	287.5	1702	0	100	400	400
##	3	2280	400	10800	9300	385	232.0	1861	100	100	400	100
##	4	2140	600	25800	11000	308	321.5	1944	100	0	300	100
##	5	1870	800	36000	17500	315	344.0	1702	0	0	300	200
##	6	1810	600	22500	10500	276	346.0	2022	100	0	300	100
##	7	1430	800	36000	24500	321	357.0	1584	0	0	300	400
##	8	2440	400	14670	6200	369	319.0	2000	100	0	400	200
##	9	2280	400	14080	9500	392	315.0	2290	100	0	400	200
##	10	1920	600	16760	12300	392	344.0	1830	100	0	400	400
##	11	1780	600	16760	12300	392	344.0	1890	100	0	400	400
##	12	1640	800	27580	18000	307	407.0	1740	0	0	300	300
##	13	1730	800	27580	18000	307	373.0	1760	0	0	300	300
##	14	1520	800	27580	18000	307	378.0	1800	0	0	300	300
##	15	1040	800	47200	20500	293	525.0	1798	0	0	300	400
##	16	1040	800	46000	21500	300	542.4	1782	0	0	300	400
##	17	1470	800	44000	23000	323	534.5	1742	0	0	300	400
##	18	3240	400	7870	6600	408	220.0	1947	100	100	400	100
##	19	3040	400	7570	5200	493	161.5	1852	100	100	400	200
##	20	3390	400	7110	6500	422	183.5	1990	100	100	400	100
##	21	2150	400	12010	9700	370	246.5	2001	100	0	300	100
##	22	1550	800	31800	15000	276	352.0	1687	0	0	300	200

modify_if

Using `modify_if()`, we can specify what columns to modify

```
ufo1 <- read_csv("https://sisbid.github.io/Data-Wrangling/data/ufo/ufo_slice_1.csv")
```

```
ufo1 %>%  
  modify_if(is.character, toupper) %>%  
  head(3)
```

```
## # A tibble: 3 × 11  
##   datetime city state country shape `duration (seconds)` `duration (hours/min)`  
##   <chr>    <chr> <chr> <chr>   <chr>          <dbl> <chr>  
## 1 7/28/20... NEW ... NY    US     CHAN...      120 2 MINS  
## 2 8/15/19... MEND... NJ    US     FIRE...       30 30 SECONDS  
## 3 10/16/2... CHAR... MI    <NA>    CIGAR       30 30 SECONDS  
## # i 4 more variables: comments <chr>, `date posted` <chr>, latitude <dbl>,  
## #   longitude <dbl>
```

Speed test! modify_if vs across

```
system.time(ufo1 %>%  
  modify_if(is.character, toupper))
```

```
##      user  system elapsed  
##         0         0         0
```

```
system.time(ufo1 %>%  
  mutate(across(where(is.character), toupper)))
```

```
##      user  system elapsed  
## 0.002    0.000    0.002
```

Iterative filtering with `if_all`

Previously we filtered for patterns or conditions..

Dilemma: Seems a bit repetitive!

```
mtcars %>%  
  filter(cyl > 3 & cyl < 8,  
         gear > 3 & gear < 8,  
         carb > 3 & carb < 8)
```

##		mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
##	Mazda RX4	21.0	6	160.0	110	3.90	2.620	16.46	0	1	4	4
##	Mazda RX4 Wag	21.0	6	160.0	110	3.90	2.875	17.02	0	1	4	4
##	Merc 280	19.2	6	167.6	123	3.92	3.440	18.30	1	0	4	4
##	Merc 280C	17.8	6	167.6	123	3.92	3.440	18.90	1	0	4	4
##	Ferrari Dino	19.7	6	145.0	175	3.62	2.770	15.50	0	1	5	6

Now we can filter multiple columns!

`if_all()`: helps us filter on multiple similar conditions simultaneously!

```
mtcars %>%  
  filter(if_all(c(cyl, gear, carb), ~.x > 3 & .x < 8))
```

##		mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
##	Mazda RX4	21.0	6	160.0	110	3.90	2.620	16.46	0	1	4	4
##	Mazda RX4 Wag	21.0	6	160.0	110	3.90	2.875	17.02	0	1	4	4
##	Merc 280	19.2	6	167.6	123	3.92	3.440	18.30	1	0	4	4
##	Merc 280C	17.8	6	167.6	123	3.92	3.440	18.90	1	0	4	4
##	Ferrari Dino	19.7	6	145.0	175	3.62	2.770	15.50	0	1	5	6

Lists

What is a list?

- Lists are the most flexible/"generic" data class in R
- Can be created using `list()`
- Can hold vectors, strings, matrices, models, list of other lists, lists upon lists!

```
mylist <- list(  
  letters = c("A", "b", "c"),  
  numbers = 1:3,  
  matrix(1:25, ncol = 5),  
  iris  
)
```

List Structure

```
head(mylist)
```

```
## $letters
## [1] "A" "b" "c"
##
## $numbers
## [1] 1 2 3
##
## [[3]]
##      [,1] [,2] [,3] [,4] [,5]
## [1,]    1    6   11   16   21
## [2,]    2    7   12   17   22
## [3,]    3    8   13   18   23
## [4,]    4    9   14   19   24
## [5,]    5   10   15   20   25
##
## [[4]]
##      Sepal.Length Sepal.Width Petal.Length Petal.Width Species
## 1           5.1         3.5         1.4         0.2      setosa
## 2           4.9         3.0         1.4         0.2      setosa
## 3           4.7         3.2         1.3         0.2      setosa
## 4           4.6         3.1         1.5         0.2      setosa
## 5           5.0         3.6         1.4         0.2      setosa
## 6           5.4         3.9         1.7         0.4      setosa
## 7           4.6         3.4         1.4         0.3      setosa
## 8           5.0         3.4         1.5         0.2      setosa
## 9           4.4         2.9         1.4         0.2      setosa
## 10          4.9         3.1         1.5         0.1      setosa
## 11          5.4         3.7         1.5         0.2      setosa
```

List referencing

```
mylist[1] # returns a list
```

```
## $letters  
## [1] "A" "b" "c"
```

```
mylist["letters"] # returns a list
```

```
## $letters  
## [1] "A" "b" "c"
```


List referencing

```
mylist[[1]] # returns the vector 'letters'
```

```
## [1] "A" "b" "c"
```

```
mylist[["letters"]] # returns the vector 'letters'
```

```
## [1] "A" "b" "c"
```

Why do this at all?

https://jennybc.github.io/purrr-tutorial/bk01_base-functions.html

You need a way to iterate in R in a data-structure-informed way. What does that mean?

You can **iterate over elements of a list**! This might be over sub data frames or different files.

Cleaning up multiple datasets

Store datasets in a list!

```
ufo1 <- read_csv("https://sisbid.github.io/Data-Wrangling/data/ufo/ufo_slice_1.csv")  
ufo2 <- read_delim("https://sisbid.github.io/Data-Wrangling/data/ufo/ufo_slice_2.tsv")  
ufo3 <- read_delim("https://sisbid.github.io/Data-Wrangling/data/ufo/ufo_slice_3.csv", delim = ":")
```

```
ufo_datasets <- list(ufo1, ufo2, ufo3)
```

Cleaning up multiple datasets

Clean names on all datasets at once:

```
library(janitor)

ufo_datasets_clean <-
  ufo_datasets %>%
  map(clean_names)
```

Cleaning up multiple datasets

Confirm columns have been cleaned! Look at `ufo_datasets_clean` or look at each with indexing, eg.:

```
ufo_datasets_clean[[1]]
```

```
## # A tibble: 20 × 11
##   datetime      city state country shape duration_seconds duration_hours_min
##   <chr>         <chr> <chr> <chr>   <chr>      <dbl> <chr>
## 1 7/28/2002 20:10 new ... ny    us    chan...    120 2 mins
## 2 8/15/1988 21:30 mend... nj    us    fire...     30 30 seconds
## 3 10/16/2010 22:... char... mi    <NA>    cigar      30 30 seconds
## 4 4/25/2013 05:30 fran... wi    us    sphe...    360 6 minutes
## 5 1/25/2012 18:22 colo... co    us    cigar     180 3 min.
## 6 3/10/2010 00:23 huson mt    us    circ...    120 several minutes
## 7 8/16/2003 17:00 hous... tx    us    form...   2700 45 minutes
## 8 4/29/2011 20:45 san ... ca    us    fire...     600 5-10 mins
## 9 6/30/2004 13:00 char... sc    us    rect...   3600 less than 1 hour
## 10 3/1/2009 14:23 huds... ny    us    sphe...     15 15 seconds
## 11 3/13/2000 20:00 san ... ca    us    light      30 15-30 seconds
## 12 9/12/2006 15:30 moja... nv    <NA>    sphe...   1800 30min or more
## 13 9/29/2012 18:30 san ... <NA> <NA>    fire...     600 7-10 minutes
## 14 8/12/2006 00:30 murr... ky    us    chan...  14400 4hr
## 15 9/1/2011 01:55 vash... wa    us    light      900 15 minues
## 16 1/3/2013 00:43 phoe... az    us    <NA>       300 5 minutes
## 17 3/3/2000 04:00 long... fl    us    light      20 20 seconds
## 18 11/29/2013 07:... i-95... sc    <NA>    other      20 20 seconds
## 19 4/5/2014 22:30 san ... tx    us    tria...     900 15 minutes
## 20 10/24/2013 19:... lake... tn    us    light      30 30 seconds
```

Cleaning up multiple datasets

Apply an anonymous function! Creates a count **tibble** for each country

```
ufo_datasets %>%  
  map(\(x) x %>% count(country))
```

```
## [[1]]  
## # A tibble: 2 × 2  
##   country      n  
##   <chr>    <int>  
## 1 us      16  
## 2 <NA>     4  
##  
## [[2]]  
## # A tibble: 2 × 2  
##   country      n  
##   <chr>    <int>  
## 1 us      17  
## 2 <NA>     3  
##  
## [[3]]  
## # A tibble: 3 × 2  
##   country      n  
##   <chr>    <int>  
## 1 ca         1  
## 2 us        15  
## 3 <NA>        4
```

List: `group_split()` a dataset

We can create a list by splitting up a dataframe. We will use `mtcars`.

```
head(mtcars)
```

##		mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
##	Mazda RX4	21.0	6	160	110	3.90	2.620	16.46	0	1	4	4
##	Mazda RX4 Wag	21.0	6	160	110	3.90	2.875	17.02	0	1	4	4
##	Datsun 710	22.8	4	108	93	3.85	2.320	18.61	1	1	4	1
##	Hornet 4 Drive	21.4	6	258	110	3.08	3.215	19.44	1	0	3	1
##	Hornet Sportabout	18.7	8	360	175	3.15	3.440	17.02	0	0	3	2
##	Valiant	18.1	6	225	105	2.76	3.460	20.22	1	0	3	1

List: `group_split()` the dataset by `cyl`

The following creates split of data for each unique `cyl` value:

```
mtcars_split <- mtcars %>% group_by(cyl) %>% group_split()
mtcars_keys <- mtcars %>% group_by(cyl) %>% group_keys() %>% pull(cyl)
names(mtcars_split) <- mtcars_keys
glimpse(mtcars_split)
```

```
## list<tibble[,11]> [1:3]
## $ 4: tibble [11 × 11] (S3: tbl_df/tbl/data.frame)
## $ 6: tibble [7 × 11] (S3: tbl_df/tbl/data.frame)
## $ 8: tibble [14 × 11] (S3: tbl_df/tbl/data.frame)
## @ ptype: tibble [0 × 11] (S3: tbl_df/tbl/data.frame)
```


List: model on each

```
mtcars_split %>%  
  map(~lm(mpg ~ wt, data = .)) %>% # apply linear model to each  
  map(summary) %>%  
  map_dbl("r.squared")
```

```
##           4           6           8  
## 0.5086326 0.4645102 0.4229655
```

Summary

- `function(x){ }` or `\(x){ }` denotes a function.
- `across` works with `mutate` or `summarize`. First specify *what* to work on, then what to *do*. `~` and `.` will help you use function arguments.
- `map` and `modify` **apply functions**. `map` returns a list, `modify` returns the same object type.
- The `purrr` package has other useful functional programming features.
- lists can be great for storing iterative work.
- `group_split` and `group_keys` can be handy with `group_by` to create a list of subset tibbles.

https://sisbid.github.io/Data-Wrangling/14_Functional_Programming/lab/functional-program-lab.Rmd